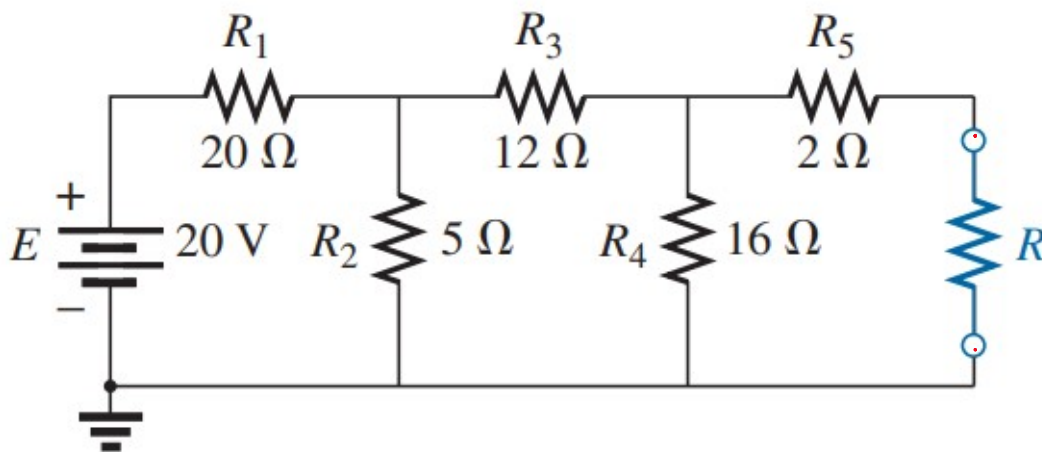

Report writing format: (Objectives, Introduction, Results& Analysis (answer the following problems), Conclusion & Discussion)

Submission Format: pdf file/word file. (1 report submission form 1 group)

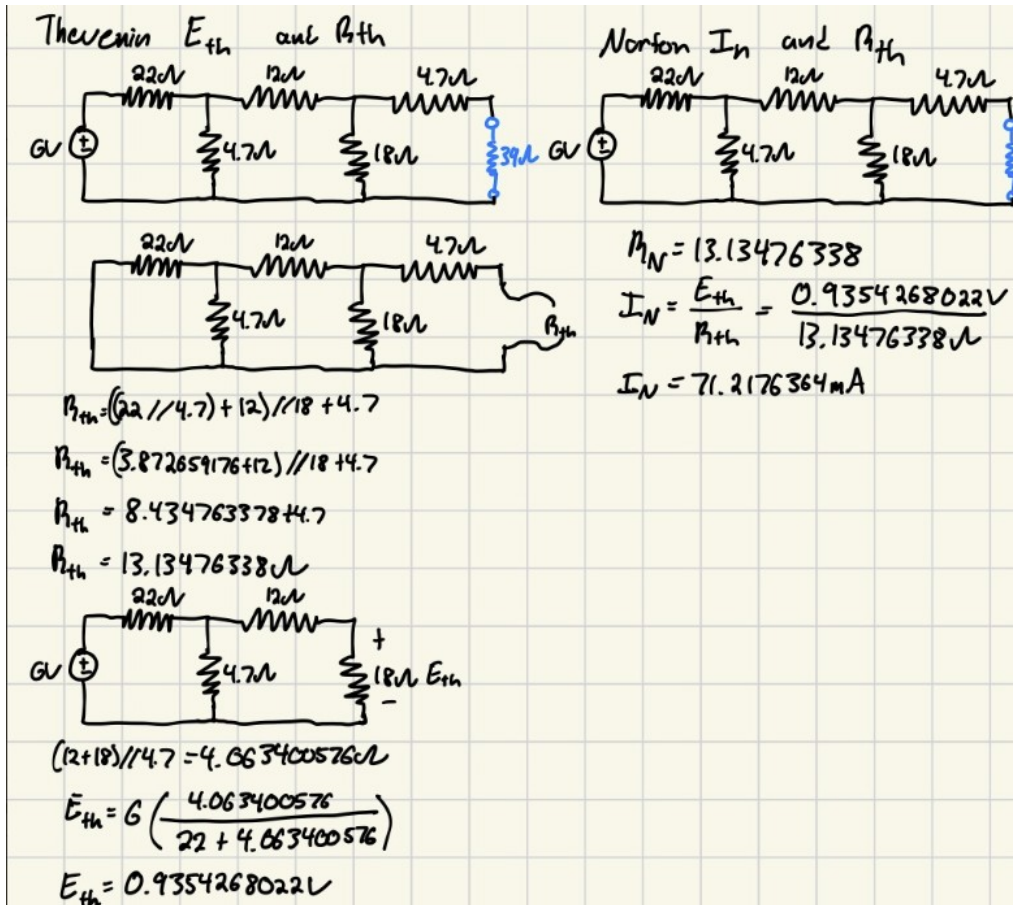
Problem Description: Proof of Thevenin, Norton, and Maximum Power Transfer theorem:

Try to select a component with a value as close as possible to the one specified in the circuit (shown below) from the available components in the laboratory. For the load resistance, you may choose any value, as long as it is well-justified. Use this chosen value consistently for all your calculations, measurements, and simulations when building the complete circuit.



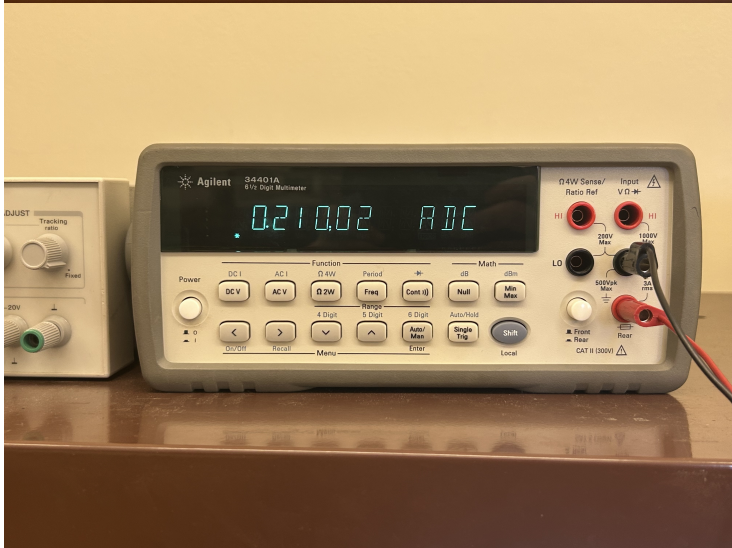
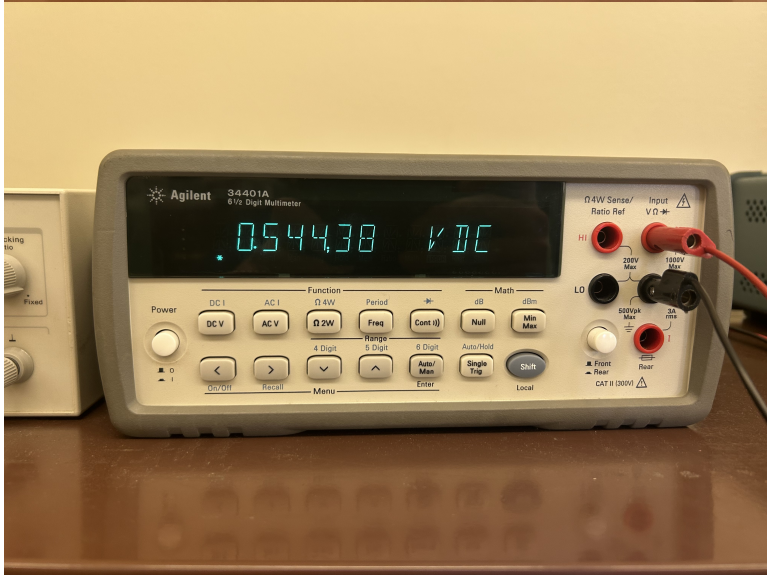
$R_1=22\Omega$, $R_2=4.7\Omega$, $R_3=12\Omega$, $R_4=18\Omega$, $R_5=4.7\Omega$, $R=39\Omega$
 $V=6V$

>Action Item 1.1: Calculation: Find the Thevenin equivalent circuit. Next, find the Norton equivalent circuit. Calculate the Norton current using the Thevenin equivalent circuit and compare it to the value obtained from the Norton equivalent circuit. Similarly, calculate the Thevenin voltage using the Norton equivalent circuit and compare it to the value from the Thevenin equivalent circuit. Do the results match?



Yes the results match from the Thevenin voltage to the Thevenin equivalent circuit.

>Action Item 1.2: Lab: Measure the Thevenin voltage, Norton current, and Thevenin/Norton resistance in the lab. Keep in mind that you will need to modify the original circuit to perform these measurements.



>Action Item 1.3: Simulation: Measure the Thevenin voltage, Norton current, and Thevenin resistance using PSpice. Ensure that you modify the original circuit as needed to obtain these values. Compile all the calculated, measured, and simulated values into Table 1 for clear comparison.

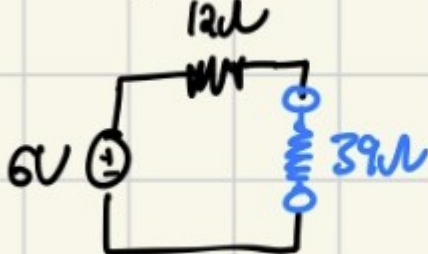
	Calculated	Measured	Simulated
R_{TH}	13.134 Ω	13.482 Ω	X
R_N	13.134 Ω	13.482 Ω	X
E_{TH}	0.93354V	0.54438V	0.561256V
I_N	0.07121A	0.21002A	0.31181A

[illegible]

>Action Item 1.4: Calculation: For the Thevenin equivalent circuit, calculate the I_L (Load Current), V_L (Load Voltage), P_L (Load Power). Organize and list all the calculated values in Table 2.

Comparison Table#2 (From calculation)	
	From Thevenin Equivalent circuit
I_L	117.6470588mA
V_L	4.588235294V
P_L	539.7923874mW

Thermin equivalent circuit



$V = IR$

$$I_L = \frac{V}{R} = \frac{6V}{12+39} = 0.1176470588A$$

$$117.6470588mA$$

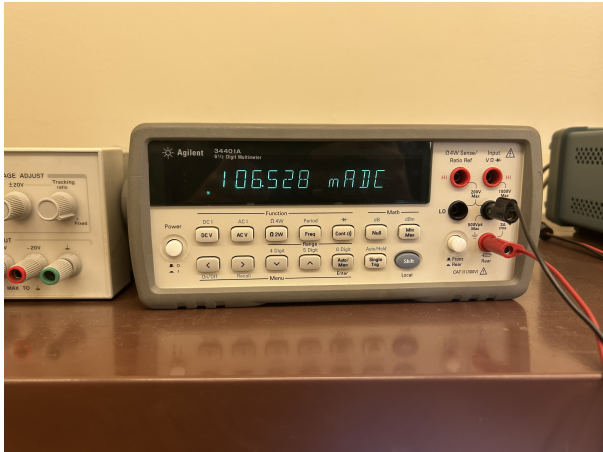
$$V_L = 6 \left(\frac{39}{12+39} \right) = 4.588235294V$$

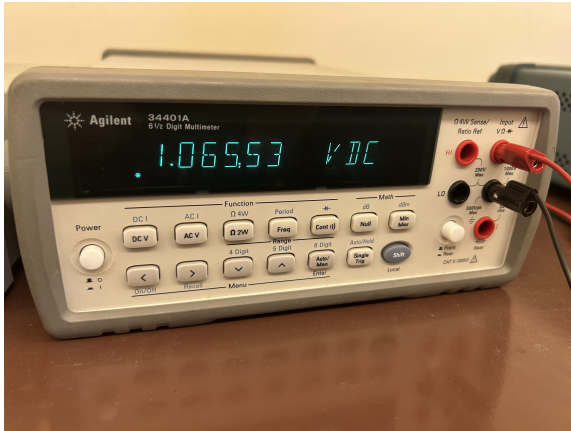
$$P_L = I_L V_L = 117.6470588mA \cdot 4.588235294V$$

$$= 539.7923874mW$$

>Action Item 1.5: Lab: Assume a resistor value and use it to construct both the original circuit and the Thevenin equivalent circuit in the lab. Measure and determine the I_L (Load Current), V_L (Load Voltage), P_L (Load Power). Record and organize all measured values in Table 3 for comparison and analysis.

Comparison Table#3 (From Lab)		
	From original circuit	From Thevenin Equivalent circuit
I_L	71.530mA	106.528mA
V_L	1.06553V	4.2864V
P_L	76.2173609mW	456.621619mW





The reasoning for the error could have been from resistor error.

Action Item 1.6: Simulation: Using the same resistor value, construct the original circuit, Thevenin equivalent, and Norton equivalent in PSpice. For each configuration, determine the I_L (Load Current), V_L (Load Voltage), P_L (Load Power). Compile and organize all the calculated and simulated values into Table 4 for clear comparison.

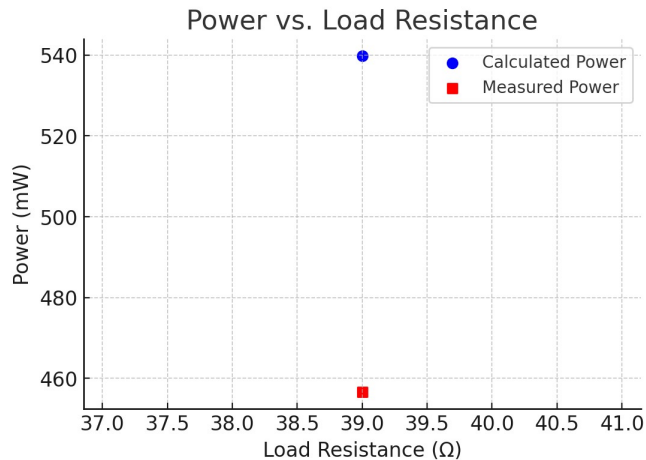
Comparison Table#4 (From simulation)			
	From original circuit	From Thevenin Equivalent circuit	From Norton Equivalent circuit
I_L	0.117647mA	0.117647mA	0.117647mA
V_L	4.5882V	4.5882V	4.5882V
P_L	539.792mW	539.792mW	539.792mW

Trace Color	Trace Name	Y1	Y2	Y1 - Y2	Y1(Cursor1) - Y2(Cursor2)	0.000			
X Values		0.000	0.000	0.000	Y1 - Y1(Cursor1)	Y2 - Y2(Cursor2)	Max Y	Min Y	Avg Y
CURSOR 1 2	V(R3:2)	561.256m	561.256m	0.000	0.000	0.000	561.256m	561.256m	561.256m

>Action Item 1.7: Lab & Calculation: For the above Thevenin equivalent circuit, Change the R_L value systematically, and calculate and measure the corresponding P_L for each value. Create a plot of R_L (x-axis) versus P_L (y-axis) to visualize the relationship between load resistance and power (calculated and measured). Determine the R_L value that results in maximum power. Verify if this value matches the prediction of the Maximum Power Transfer Theorem, which states that maximum power is transferred when $R_L = R_{TH}$ (Thevenin resistance). Use Excel to create the plot, format it appropriately, and ensure it includes labeled axes, a title, and clear markers for the maximum power point.

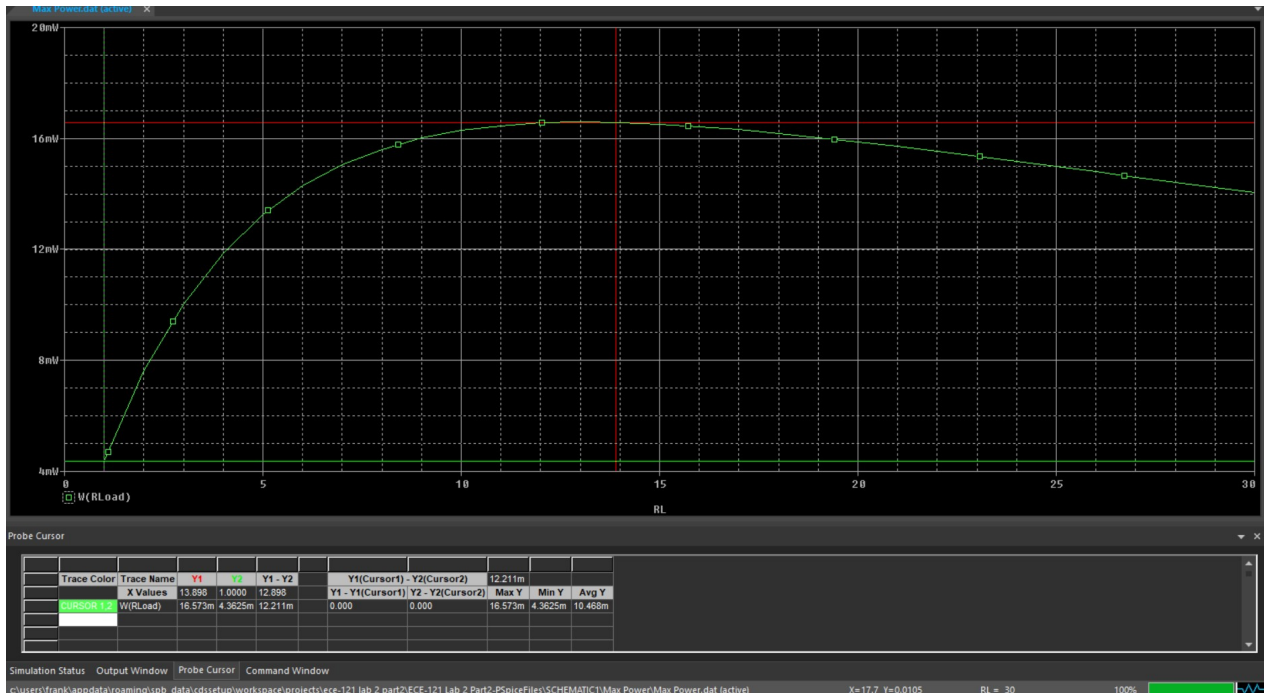
Add the plot to your report with a brief explanation of your observations and comparison with the theorem.

R_L	P_L (calculated)	P_L (measured)
39Ω	539.7923874mW	456.621619mW



As we can see from the plot there is a big difference in power between calculated and measured. This could be from component wear and tare which has caused us to get invalid results.

>Action Item 1.8: Simulation: For the above Thevenin equivalent circuit, Change the R_L value systematically and calculate the corresponding P_L for each value. Create a plot of R_L (x-axis) versus P_L (y-axis) to visualize the relationship between load resistance and power. Does it match with the maximum power transfer theorem? Add the plot screenshot to your report.



Yes, it does match with the maximum power transfer theorem because we can see that it does reach a maximum power where they intersect.

Add your all simulated screenshots and measured lab pictures as proof of your answer. For measurements show your circuit (original and equivalent), to the TA as a demo.

Grading Rubric:

Writing = 20 points

Answer = 80 points

Lab measurement, simulation, and hand calculation have an equal distribution of points.

Contribution Chart: You will add a contribution chart as shown below of your group members at the beginning of your lab report if you work in a group. You will be graded based on 20% of your contribution and 80% of your group contribution. Every group member should be knowledgeable of every part of the lab assignment.

	Frank Tamburro	
Initial Research	X	
Simulation	X	
Lab	X	
Calculation	X	
Documentation & Report writing	X	

